



LISRec: Modeling User Preferences with Learned Item Shortcuts for Sequential Recommendation

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Background & Motivation

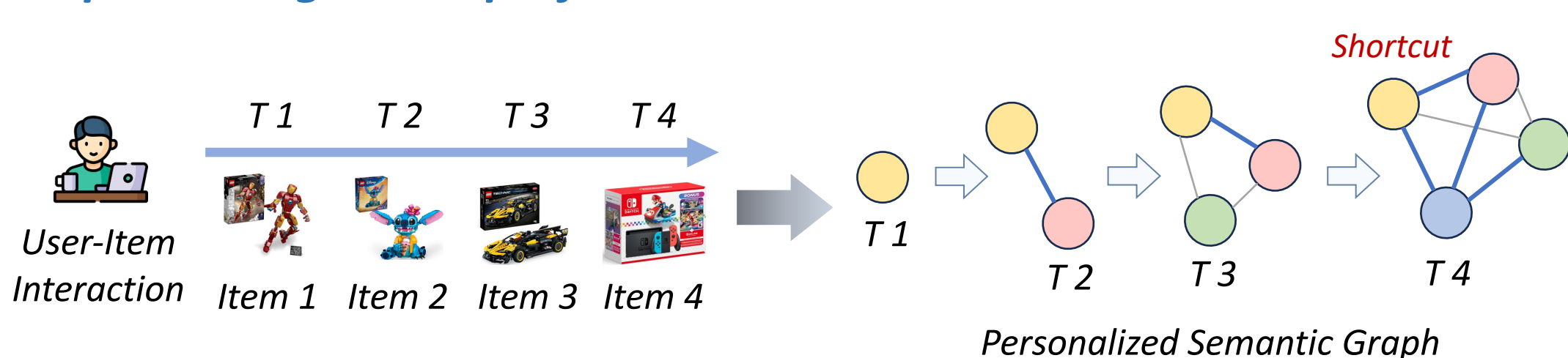
Noise in Sequential Recommendation

- User histories often contain **accidental clicks** and **transient behaviors** that **do not reflect stable preference**, leading to **suboptimal recommendations**.



Semantic Shortcuts for Denoising

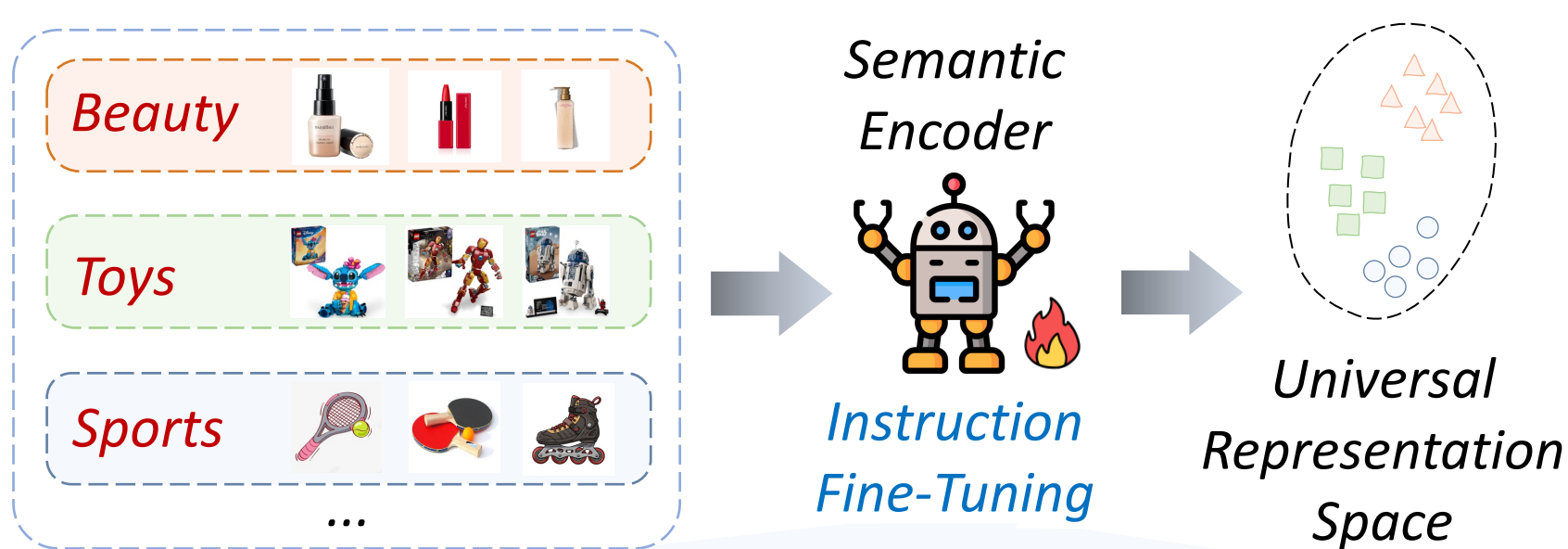
- LISRec builds a **personalized semantic graph** over the **full interaction history** and **extracts shortcut**, **filtering noise** while **preserving stable preferences**.



Extracting Shortcuts via Maximal Semantic Connectivity

Task-Agnostic Semantic Item Encoding

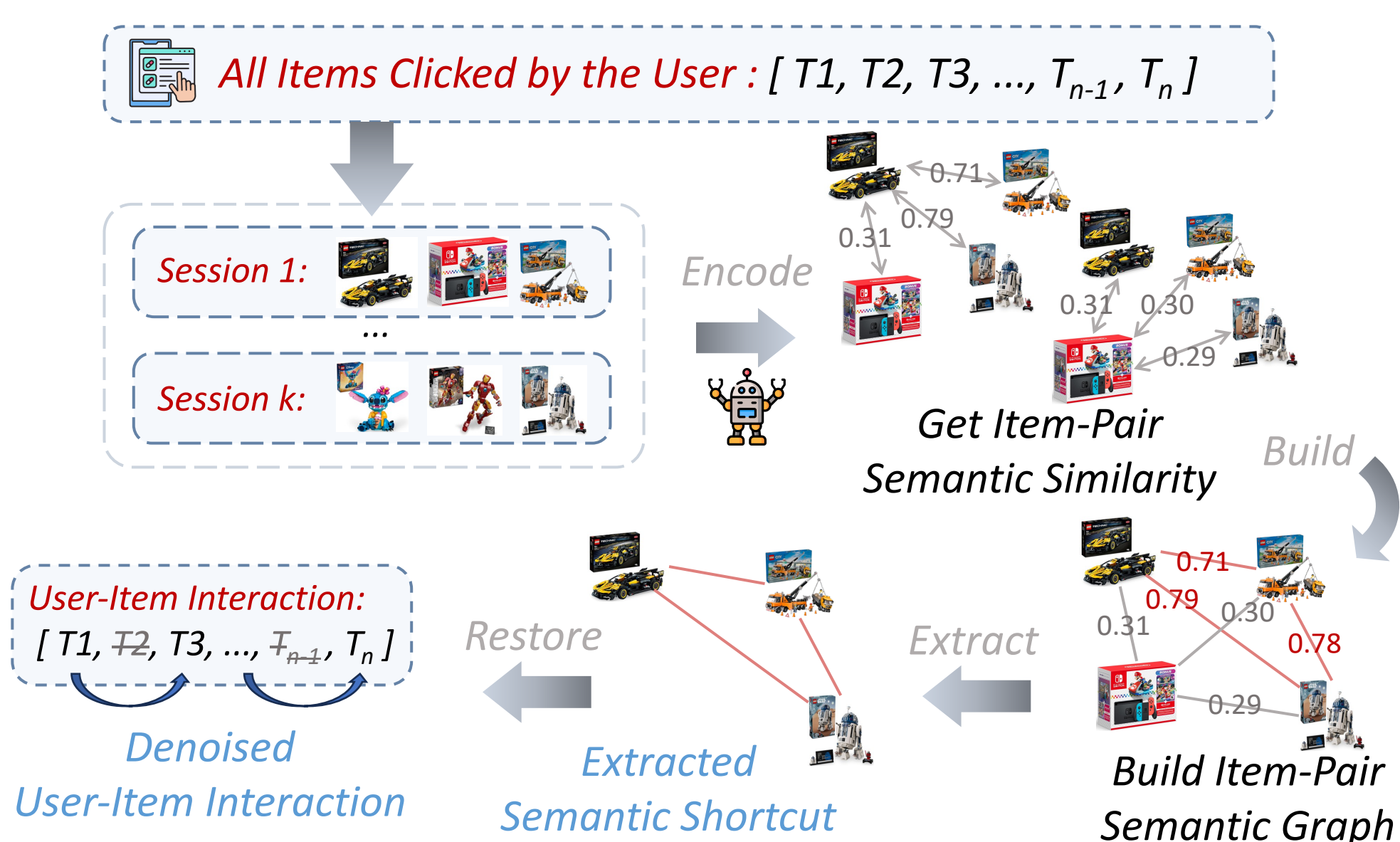
- LISRec first represents each item using **textual attributes only**, **excluding item IDs**. It then **instruction-tunes the semantic encoder** on **next-item prediction**, producing a **task-agnostic semantic space** for pairwise similarity estimation.



This is T dataset. Here is the visit history list of user:
 $\langle \text{Attr} \rangle_1: V(\langle \text{Attr} \rangle_1), \dots, \langle \text{Attr} \rangle_k: V(\langle \text{Attr} \rangle_k), \text{recommend next item}$

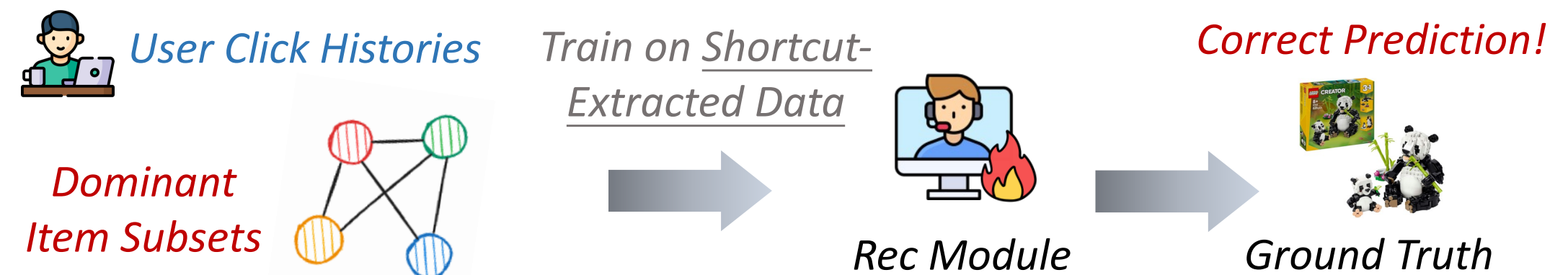
Modeling Stable Preferences with Personal Item Graphs

- LISRec collects all items clicked by each user **across sessions**. Then, it builds a **personalized item graph** by linking item pairs with **semantic similarity**. Finally, LISRec extract the **largest connected component** as the **user's semantic shortcut**.



Personalized Recommendation Based on Semantic Shortcuts

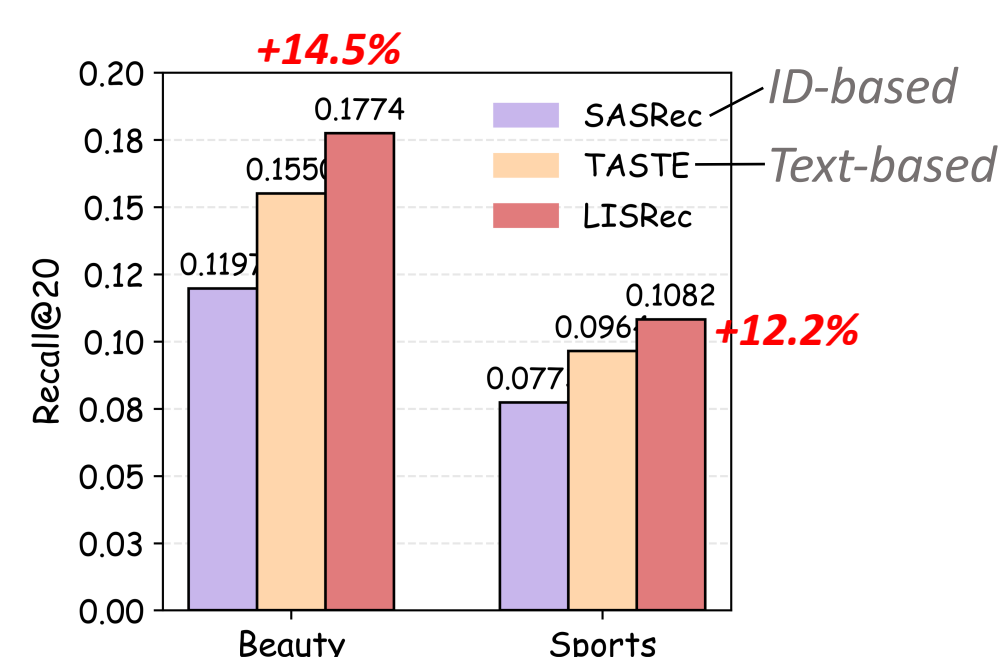
- Extracting **dominant item subsets** from user click histories to learn **semantic shortcuts** for next-item prediction.



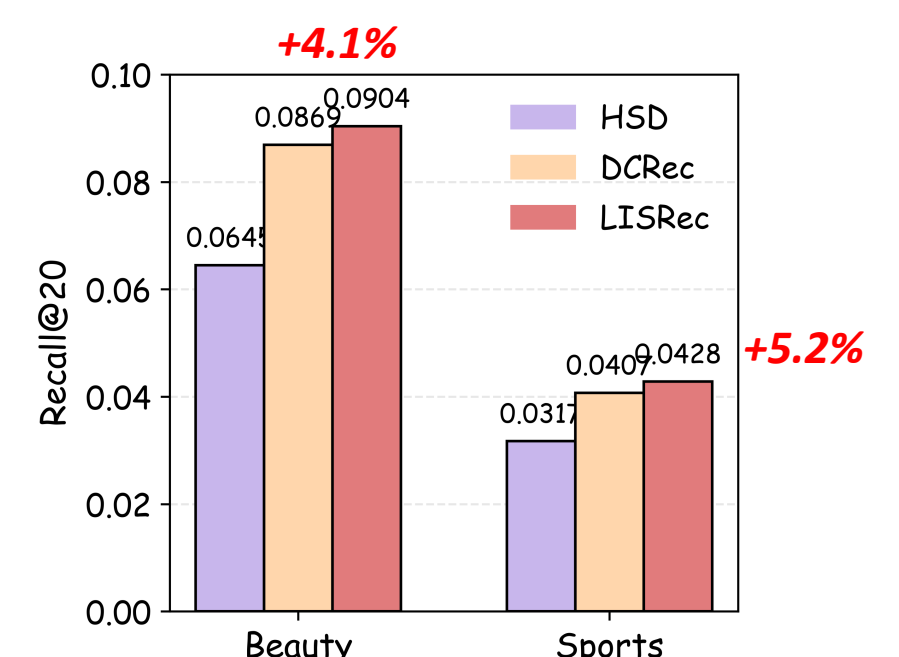
Experimental Results & Analysis

Overall Performance

- LISRec **outperforms all baseline models** across scenarios, achieving the SOTA performance on both **Text-based model (Liu 2023, TASTE)** and **ID-based model (Kang 2018, SASRec)**, while performing well on a series of SOTA **Denoising Frameworks**.



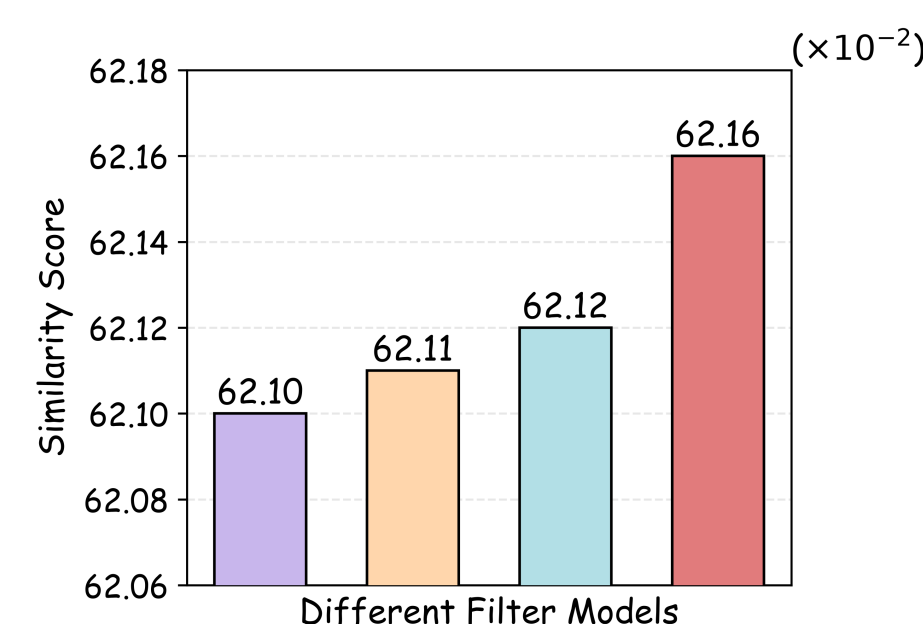
Recommendation Performance.
(LISRec using TASTE backbone)



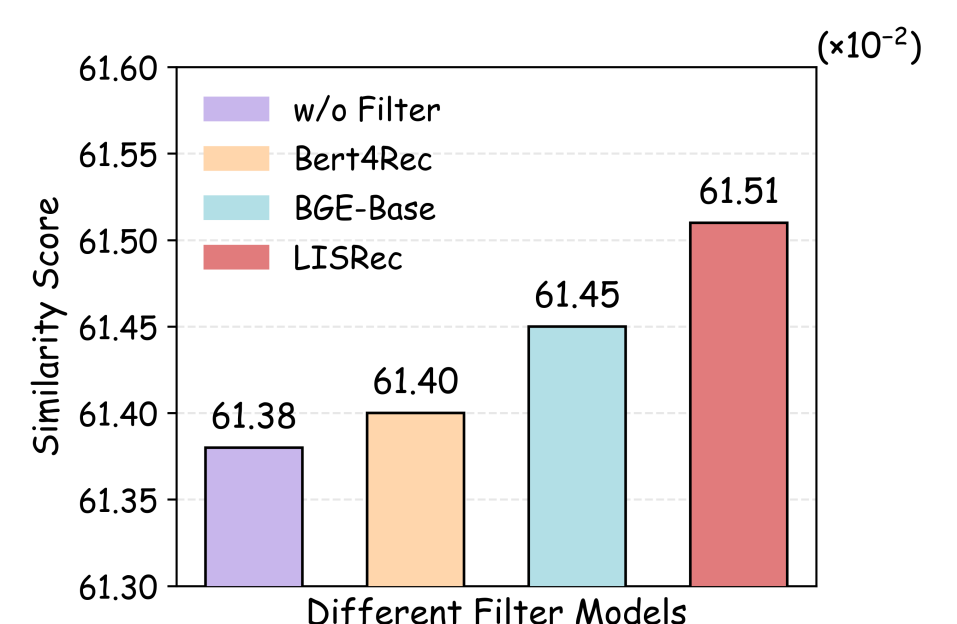
Denoising Performance.
(LISRec using Bert4Rec backbone)

Item Representations for Shortcut Extraction

- LISRec **learns item representations** that are better suited for **shortcut extraction**, while better **identifies interactions** that truly reflect **stable user preferences**.



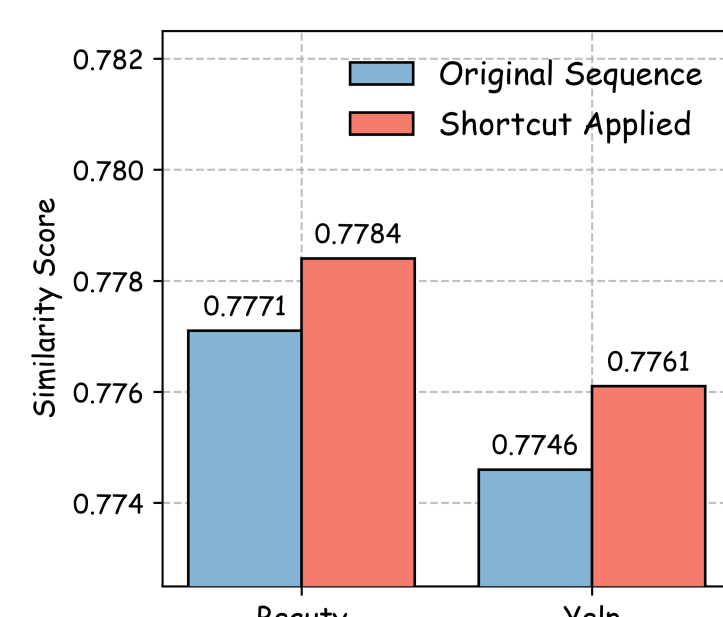
Retained-Item Coherence.



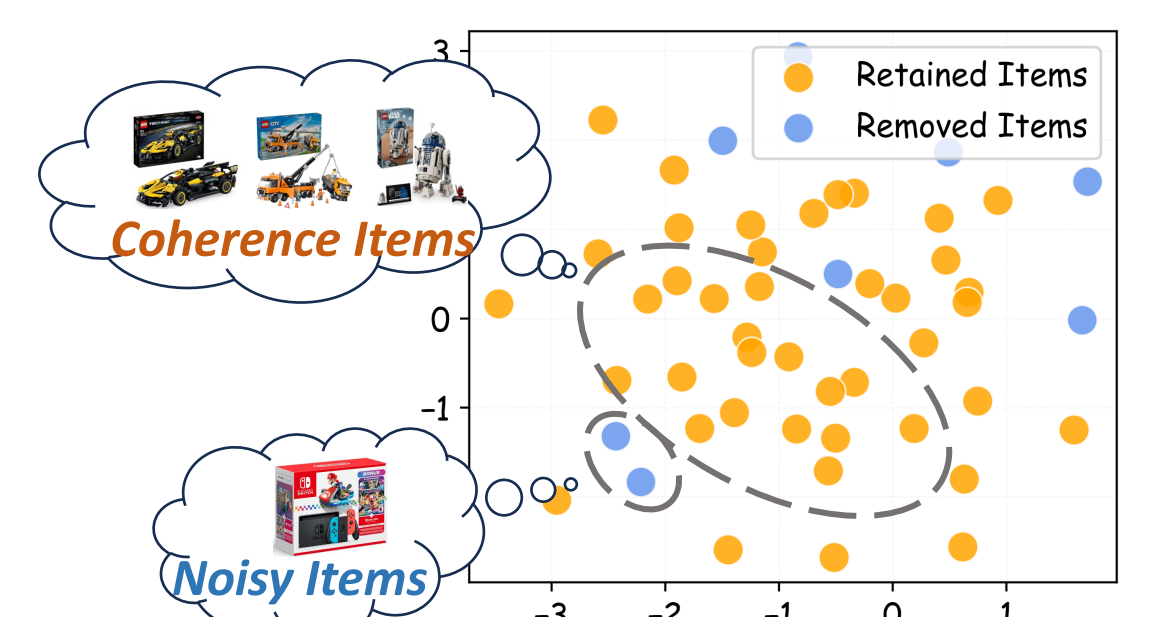
Shortcut-Target Similarity.

Shortcut Extraction Enhances the Recommendation Module

- Shortcut extraction strengthens **user-target alignment**, bringing user representations **closer to target-item embeddings**.
- LISRec separates removed items from retained ones, yielding a **cleaner and more discriminative representation space**.



Representation Alignment of User Seq. and Target Item.



t-SNE Separation of Retained Items and Removed Items.

Resources & Contact



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